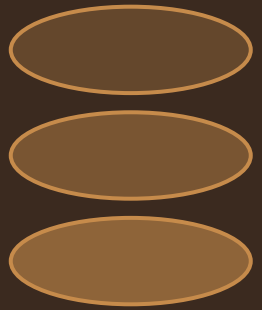
MSE 434

GROUP 5
FINAL PRESENTATION

JULY 7, 2026



Nespresso Coffee Pods Supply Chain

Balancing cost, service, and stockout risk across Nespresso's North American distribution network.

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Scope & Decision Question

DECISION QUESTION

Should Nespresso's North American distribution network remain two centralized distribution centers on ocean freight, or move to one of two alternative configurations that better balance inventory, freight, and facility costs against service level and stockout risk?

The project analyzes the North America segment of Nespresso's coffee capsule supply chain, beginning at the point capsules arrive in North America and ending with delivery to the end customer.

IN SCOPE

- Inventory positioning & safety stock policy at NA DC(s)
- Inbound freight mode into NA (ocean vs. air, trigger logic)
- NA network configuration (2-DC vs. single centralized DC)
- Cost, service, and risk tradeoffs across these decisions

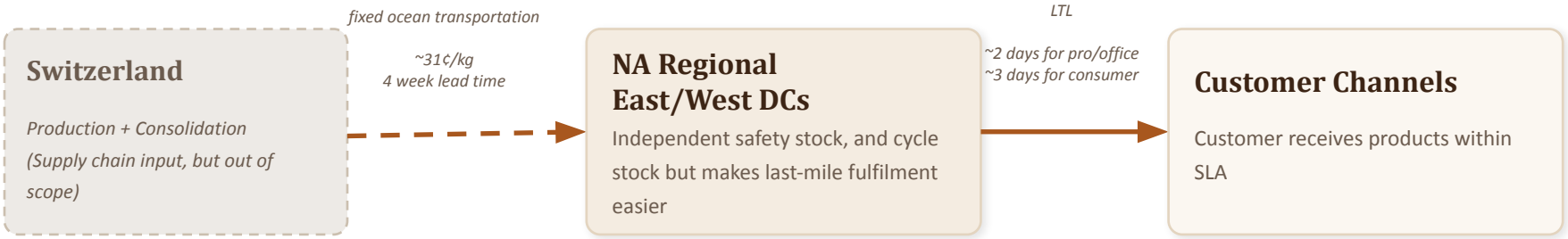
OUT OF SCOPE

- Coffee sourcing, farming, and Swiss production operations
- Swiss manufacturing scheduling or facility investment
- Transoceanic mode/route decisions, held fixed as input
- Sustainability/recycling programs; markets outside NA

EVALUATION METRICS

Service: CSL, fill rate, stockout frequency **Risk:** demand CoV (σ/μ) **Cost:** cycle stock, safety stock, inbound freight, facility

Current State Supply Chain and Pain Points



Issues under Current State Supply Chain

Aggregated Inventory

Increased safety stock, leading to high holding costs

Multi-week Replenishment Risk

Both DCs are open to long lead times, with no way of quickly replenishing stock in the event of a stockout

Volatile Coffee Supply

Coffee supply chains have been volatile, with prices and shortages fluctuating over past years

MAIN PAIN POINT (NA-FOCUSED)

Since 2007, NA demand has run on two independent DCs, each with separate safety stock and ocean freight from Switzerland. This inflates buffer costs and concentrates service risk, with no way to protect against upstream supply shortages once they hit.

PERFORMANCE DIAGNOSIS

Verified Data and Open Assumptions

Cost

Assumptions:

- Components: cycle-stock holding + safety-stock holding + inbound freight (+ one-time facility change for Alt 3)
- Freight gap: $5.47/kg air vs. $0.23–0.40/kg ocean (~20–25x) [2]
- Holding rate: ~20%/yr of unit cost

Service

Verified:

- NA delivery SLA: 2 days (Pro/office), 3 days (consumer) [6][8]

Assumptions:

- Swiss–NA replenishment lead time: 3–5 weeks
- Cycle Service Level (CSL): target no-stockout probability, set per alternative

Risk

Verified:

- Arabica price swing: $4.30 → $2.80/lb (≈35%) [5]
- Global coffee deficit: 17.91M bags, CY 2021/22–2024/25 [3]

Assumptions:

- NA weekly demand variability: CV ~15–25% (category proxy)

Data status: Verified figures (SLA, Arabica swing, coffee deficit) are sourced and cited; freight gap, holding rate, lead time, and demand variability remain team working assumptions.

ANALYSIS APPROACH

Analysis Using Course Tools

Three tools from the course apply directly:

Risk pooling (location models), mode selection (transportation), safety stock sizing (inventory). Each is drawn from course formulas and applied to the verified/assumed Evidence Base data.

Tool 1: Risk Pooling

SS $\propto \sqrt{n}$: consolidating from 2 DCs to 1 cuts total safety stock by about 29%. This is the basis for Alt 3's pooling edge, but only if east/west demand isn't strongly correlated.

Tool 2: Mode Selection

Total logistics cost = transport + pipeline + safety stock. Air's premium can be offset by its shorter lead time on triggered volume, though this isn't quantified yet.

Tool 3: Safety Stock Sizing

SS = $k\sigma_L$: buffer scales with the service target (k) and lead-time uncertainty (σ_L). Air's shorter lead time lowers σ_L on Alt 2's buffer, pending a set P_1 target.

Open inputs: NA demand volume, unit value, air lead time, and the target service level (P_1) aren't set yet. These are needed to turn these relationships into dollar figures.

WORKED EXAMPLE

Sizing Safety Stock: A Worked Example

Based on example figures:

NA demand volume and CV aren't confirmed yet, so this walks through the formulas using example figures, 10,000 units/wk demand, 20% CV, and a 4-wk ocean lead time, to show how Tool 1 and Tool 3 apply in practice.

Step 1: Lead-Time Uncertainty

$\sigma_{weekly} = CV \times \text{mean demand} = 0.20 \times 10{,}000 = 2{,}000$ units. $\sigma_L = \sigma_{weekly} \times \sqrt{L} = 2{,}000 \times \sqrt{4} = 4{,}000$ units.

Step 2: Safety Stock (95% Target)

$SS = k \times \sigma_L = 1.645 \times 4{,}000 \approx 6{,}580$ units for a single pooled DC at a 95% service level.

Step 3: Two-DC Comparison

Splitting the same demand across 2 independent DCs needs $\sqrt{2} \times 6{,}580 \approx 9{,}306$ units combined, about 2,730 more than pooled.

What this shows: pooling into one DC needs about 29% less safety stock than splitting into two, for the same service level, matching the Tool 1 result. Real Nespresso figures would change the numbers, not this relationship.

ALTERNATIVES CONSIDERED

Three Configurations

Each holds Swiss production and ocean lead time fixed, differing only in NA network structure and replenishment policy.

Alt 1: Status Quo

Two-DC · Ocean-Only

East/West DCs, each on ocean freight with independent safety stock. Current baseline per 2007 SEC filing on Nespresso USA's bi-coastal network.

COST

Baseline cycle + safety stock; ocean freight per DC; no facility cost

KEY LIMITATION

2-node coverage at today's scale unverified; loses pooling benefit

Alt 2: Elevated Service

Two-DC + Air Trigger

Same network, plus a reorder trigger: air freight kicks in below a threshold. Mechanism mirrors Zara's ocean/air split and apparel brands' stockout-driven air use.

COST

Same cycle stock; reduced safety stock; ocean + air premium on triggered volume

KEY LIMITATION

Assumes reliable air capacity at both coasts; premium is estimated, not quoted

Alt 3: Consolidation

Single Centralized DC

One NA DC, pooled safety stock. Precedent by Zara's single-hub model.

COST

Pooled safety stock (lower, if uncorrelated); one consolidated ocean shipment

KEY LIMITATION

Reverses Nespresso's own 2007 bi-coastal rationale; longer cross-coast lead time

Implementation and Rollout

Alternative 2, Elevated Service (Bi-Coastal Two-DC):

Nespresso should implement the elevated service alternative with bi-coastal distribution centres and and an air freight trigger. The solution preserves the strengths of the current network while adding operational flexibility through targeted use of air freight rather than requiring costly structural charges.

Air Capacity

- Not yet confirmed with a carrier
- **Risk:** correlated demand spikes push both DCs below threshold at once, forcing sites to compete for limited capacity
- **Fix:** standing carrier agreement before rollout

Freight Rate

- Current 20–25x gap is an industry benchmark, not a Nespresso-quoted rate
- **Risk:** Recommendations cost case depends on this premium staying manageable
- **Fix:** validate against a real carrier quote, first priority, fastest to resolve

Inventory Systems

- Trigger needs real-time breach detection; current tracking is periodic
- **Risk:** lag between a breach and a triggered shipment
- **Fix:** confirm what current systems can detect before finalizing trigger design

Sequencing: Validate freight rate → confirm air capacity → evaluate system upgrades. None require a change to the two-DC structure. Resolving these three dependencies is what moves the recommendation from preliminary to implementation-ready

LIMITATIONS & SENSITIVITY

What Could Change the Answer

The analysis works well, only two hypotheticals would shift the outcome:

If East/West demand correlation or the air-freight numbers turned out different than assumed, the ranking could shift. Everything else only changes the size of the gap, not the winner.

Demand Correlation

Alternative 3's edge assumes East/West demand move independently. If they correlate, the edge shrinks.

Freight & Lead Time

Alternative 2's premium and lead time are benchmarks, not quoted rates. A higher premium narrows the case.

Everything Else

Demand volume, unit value and holding rate shift cost size, not who wins, since all alternatives move together.

Why just these two: demand correlation and the air premium/lead time change the shape of the comparison. Every other input just changes its scale, not the outcome.

Citations

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